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Prof. Rao

Senior Project

Weekly Report 4 — 11/10/2026

This week I have been getting a lot more coding done — I've been focusing on figuring out my algorithms and making them work with the GUI library that I have. More specifically, I now have exposure working as a slider, and during this process I realized how important optimisation is, especially with the nature of my project. This is why I have been working on exactly that — developing a downscaling algorithm that will work as follows: because the slider works in a weird way — it applies the function every time the slider moves, i.e. every small movement is applied, it really quickly starts to lag behind the cursor, I decided to make it so that whenever the cursor holds and moves the slider, I will use a downscaled version of the image, to serve as a preview to the change made. In order to do this I am going to have multiple algorithms and I will compare how efficient they are and how well they serve my purpose — for now I have 2: 1 which takes a neighbourhood of 4 pixels and averages out their colors and a more complex one — imagine scaling the downscaled image and taking the location of every 'downscaled' pixel. Take some area of pixels around it and create an approximation for the values of the downscaled pixel. This will result in the ability to scale down to any desired size, even if not proportional to the original image, and should be more efficient and just better. Practice will show the rest.

For next week, I will be focusing on writing the thesis and getting ready for the midterm preparation, because I have been slacking in that aspect.

GitHub repo link: https://github.com/emolindar/SeniorProjectPemo